

Boolean Algebra

Def.: A non-empty set B with two binary operations $+$ and $\times$, a unary operation $'$, and two distinct elements 0 and 1 is called a Boolean Algebra, denoted by $(B, +, \times, ', 0, 1)$ if and only if the following properties are satisfied.

Axioms of Boolean Algebra

1. If $a, b, c \in B$, then

1. Commutative laws

$$(a) \quad a + b = b + a$$

$$(b) \quad a \times b = b \times a$$

2. Distributive laws

$$(a) \quad a + (b \times c) = (a + b) \times (a + c) \quad (b) \quad a \times (b + c) = (a \times b) + (a \times c)$$

3. Identity laws

$$(a) \quad a + 0 = a$$

$$(b) \quad a \times 1 = a$$

4. Complement laws

$$(a) \quad a + a' = 1$$

$$(b) \quad a \times a' = 0$$

Basic Theorem

Let $a, b, c \in B$ then

1. Idempotent laws

$$(a) \quad a + a = a$$

$$(b) \quad a \times a = a$$

$$\text{R.H.S.} = a$$

$$= a + 0 \quad \text{by identity law}$$

$$= a + (a \times a') \quad \text{using complement law}$$

$$= (a + a) \times (a + a') \quad \text{using distributive law}$$

$$= (a + a) \times 1 \quad \text{using complement law}$$

$$= a + a \quad \text{using identity law.}$$

Notes

(b) L.H.S. = $a \cdot a$

$$= a \cdot a + 0$$

By identity law

$$= a \cdot a + a \cdot a'$$

Using Complement law

$$= a \cdot (a + a')$$

Using Distributive law

$$= a \cdot 1$$

Using Complement law

$$= a = R.H.S. //$$

Using identity law.

2. Boundedness law

(a) $a + 1 = 1$

(b) $a \cdot 0 = 0$

3. Absorption laws

(a) $a + (a \cdot b) = a$

(b) $a \cdot (a + b) = a$

4. Associative laws

(a) $(a + b) + c = a + (b + c)$

(b) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

Proof of Boundedness law

2(a) $a + 1 = 1$

Let R.H.S. = 1

$$= a + a'$$

by complement law

$$= a + a' \cdot 1$$

using identity law

$$= (a + a') \cdot (a + 1)$$

Using distributive law

$$= 1 \cdot (a + 1)$$

Using complement law

$$= (a + 1) \cdot 1$$

" commutative law

$$= a + 1 = L.H.S. //$$

$$(b) a \cdot 0 = 0$$

$$R.H.S. = 0$$

$$= a \cdot a'$$

$$= a \cdot (a' + 0)$$

$$= (a \cdot a') + (a \cdot 0)$$

$$= 0 + (a \cdot 0)$$

$$= (a \cdot 0) + 0$$

$$= a \cdot 0 \quad \therefore L.H.S. //$$

Proof of Absorption laws

$$3(a) a + (a \cdot b) = a$$

$$L.H.S. = a + (a \cdot b)$$

$$= a \cdot 1 + a \cdot b$$

by Identity law

$$= a \cdot (1 + b)$$

Using Distributive law

$$= a \cdot (b + 1)$$

Using commutative law

$$= a \cdot 1$$

Using boundedness law

$$= a = R.H.S. //$$

Using identity law

$$(b) a \cdot (a + b) = a$$

$$L.H.S. = a \cdot (a + b)$$

$$= (a + 0) \cdot (a + b)$$

by Identity law

$$= (a + 0 \cdot b)$$

Using distributive law

$$= a + (b \cdot 0)$$

Using commutative law

$$= a + 0$$

Using boundedness law

$$= a$$

Using identity law

$$= R.H.S. //$$

Notes

Proof of Associative law

$$4(a) \quad (a+b)+c = a+(b+c)$$

$$R.H.S. = a+(b+c)$$

$$= [a+(b+c)] \cdot 1 \quad \text{By Identity law}$$

$$= [a+(b+c)] \cdot (c+c') \quad \text{By complement law}$$

$$= [a+(b+c)] \cdot c + [a+(b+c)] \cdot c'$$

$$= [c \cdot \{a+(b+c)\}] + [c' \cdot \{a+(b+c)\}]$$

$$= [c \cdot a + c \cdot (b+c)] + [c' \cdot a + c' \cdot (b+c)]$$

$$= [c \cdot a + c] + [c' \cdot a + c' \cdot (b+c)]$$

Using Absorption law

$$= c + [c' \cdot a + c' \cdot (b+c)] \quad \text{" "}$$

$$= c + [c' \cdot a + c' \cdot b + c' \cdot c]$$

$$= c + [c' \cdot a + c' \cdot b + 0]$$

$$= c + [c' \cdot a + c' \cdot b]$$

$$= c + [c' \cdot (a+b)]$$

$$= (c+c') \cdot [c+(a+b)]$$

$$= 1 \cdot [c+(a+b)]$$

$$= c+(a+b)$$

$$= (a+b)+c = L.H.S. \quad \checkmark$$

$$(b) (a \cdot b) \cdot c = \{a \cdot (b \cdot c)\}$$

$$R.H.S. = a \cdot (b \cdot c)$$

$$= [a \cdot (b \cdot c)] + 0 \quad \text{by Identity law}$$

$$= [a \cdot (b \cdot c)] + (c \cdot c')$$

$$= \{[a \cdot (b \cdot c)] + c\} \cdot \{[a \cdot (b \cdot c)] + c'\} \quad \text{Using Distributive law}$$

$$= \{c + [a \cdot (b \cdot c)]\} \cdot \{c' + [a \cdot (b \cdot c)]\} \quad \text{Using Comm. law}$$

$$= \{(c + a) \cdot (c + (b \cdot c))\} \cdot \{(c' + a) \cdot (c' + (b \cdot c))\}$$

$$= \{(c + a) \cdot c\} \cdot \{(c' + a) \cdot (c' + (b \cdot c))\}$$

$$= \{c \cdot (c + a)\} \cdot \{(c' + a) \cdot (c' + (b \cdot c))\} \quad \text{Using Comm & absorption law}$$

$$= c \cdot \{(c' + a) \cdot (c' + (b \cdot c))\} \quad \text{Using absorption law}$$

$$= c \cdot \{(c' + a) \cdot ((c' + b) \cdot (c' + c))\}$$

$$= c \cdot \{(c' + a) \cdot ((c' + b) \cdot 1)\}$$

$$= c \cdot \{(c' + a) \cdot (c' + b)\}$$

$$= c \cdot [c' + (a \cdot b)]$$

$$= (c \cdot c') + c \cdot (a \cdot b)$$

$$= 0 + c \cdot (a \cdot b)$$

$$= c \cdot (a \cdot b) + 0$$

$$= c \cdot (a \cdot b)$$

$$= (a \cdot b) \cdot c = L.H.S. \quad \checkmark$$

Notes

Uniqueness of Complement

Theorem: Prove that the complement of every element on a boolean algebra B is unique.

Proof: Consider an element $a \in B$.

Let a has two complements a' and a'' in B .

$$\left. \begin{aligned} a + a' &= a' + a = 1 \\ a \cdot a' &= a' \cdot a = 0 \end{aligned} \right\} \text{--- (i)}$$

$$\left. \begin{aligned} a + a'' &= a'' + a = 1 \\ a \cdot a'' &= a'' \cdot a = 0 \end{aligned} \right\} \text{--- (ii)}$$

We know that,

$$\begin{aligned} a' &= a' \cdot 1 \\ &= a' \cdot (a + a'') \\ &= a' \cdot a + a' \cdot a'' \\ &= 0 + a' \cdot a'' \\ &= a \cdot a'' + a' \cdot a'' \\ &= (a + a') \cdot a'' \\ &= 1 \cdot a'' \\ &= a'' \end{aligned}$$

Hence, the complement is unique. //

Involution law

Theorem: If B is a boolean algebra, then $a'' = a$
 $\forall a \in B$ where $a'' = (a')'$

Proof:

$$a'' = a'' + 0 = a'' + a \cdot a' = (a'' + a) \cdot (a'' + a')$$

$$= (a'' + a) \cdot 1 \quad \left[\begin{array}{l} \text{By complement law,} \\ a + a' = 1 \end{array} \right]$$

$$= (a'' + a) \cdot (a + a')$$

$$= (a + a'') \cdot (a + a')$$

$$= a + (a'' \cdot a')$$

$$= a + 0 = \underline{\underline{a}} \quad \text{proved.}$$